

TalentForge POC — Day-by-Day Execution Plan

Team: Rohan (Backend · Sandbox · Infra) · Karthikeyan (Frontend · Tools · Deploy) | **Start:** Mon Jul 13, 2026 → **Launch:** Fri Aug 14, 2026

■ Header ■ Day cell ■ Joint session **P0** critical path **STRETCH** skipable

WEEK 1 — Foundation, Auth & API Contract (Jul 13–17)

Day	Rohan — Backend / Sandbox / Infra	Karthikeyan — Frontend / Tools	Done When
Day 1 Mon Jul 13	- Init monorepo: <code>npm init -w backend -w frontend -w worker</code> - Scaffold <code>backend/</code>: Express + TypeScript (<code>tsx</code>), folders <code>src/routes</code>, <code>src/services</code>, <code>src/middleware</code> <code>docker-compose.yml</code>: postgres:16-alpine, redis:7-alpine, minio (S3-compatible) <code>schema.prisma</code>: User, Problem, Submission, Badge, PsychProfile, Review → <code>prisma migrate dev</code>	- Scaffold <code>frontend/</code>: Vite + React 18 + TS - Install <code>tailwindcss</code>, <code>@tanstack/react-query</code>, <code>react-router-dom</code>, <code>socket.io-client</code> - Design tokens in <code>tailwind.config.js</code> (brand palette, Inter font) - Build <code>AppShell.tsx</code> (sidebar + topbar) + Login/Register UI (<code>react-hook-form</code> + <code>zod</code>)	<code>docker compose up</code> boots PG + Redis + MinIO - Tables visible in <code>psql \dt</code> - Shell + auth pages render
Day 2 Tue Jul 14	<code>routes/auth.ts</code>: register (<code>bcrypt</code> ×12), login → JWT 15m + refresh in Redis <code>SET refresh:{id} EX 604800</code> <code>GET /auth/me</code> + <code>requireAuth</code> middleware - Google OAuth via <code>passport-google-oauth20</code> <code>zod validate(schema)</code> on all bodies	- Axios instance with JWT interceptor + auto-refresh on 401 - MSW mock server (<code>src/mocks/handlers.ts</code>) for all planned endpoints - Wire auth forms → token storage <code>RequireAuth.tsx</code> protected-route wrapper	- Register → login → <code>/auth/me</code> works end-to-end - Google OAuth round-trip OK - UI runs fully on mocks when backend is down
Day 3 Wed Jul 15	Joint 2h: freeze <code>openapi.yaml</code> (~25 endpoints) + socket events: <code>grading:queued/running/complete/failed</code> <code>services/s3.ts</code>: presigned PUT/GET (@aws-sdk/client-s3 → MinIO local, S3 prod) - Problem CRUD: <code>GET /problems?tier=</code>, <code>GET /problems/:slug</code> (hidden test cases stripped via Prisma select)	Joint 2h: contract freeze — regenerate MSW mocks from spec - Integrate <code>@monaco-editor/react</code> → <code>CodeEditor.tsx</code> - Language selector (py / js / java) loads boilerplate per problem - Theme toggle, font-size, <code>Ctrl+Enter</code> = run	<code>openapi.yaml</code> committed — both build against it - Monaco highlights all 3 languages - Boilerplate loads per problem

Day	Rohan — Backend / Sandbox / Infra	Karthikeyan — Frontend / Tools	Done When
Day 4 Thu Jul 16	<code>prisma/seed.ts</code>: 3 problems (Two Sum, LRU Cache, Rate Limiter) with 8–10 hidden test cases each - BullMQ queue <code>grading</code>; <code>POST /submissions</code> → S3 upload → enqueue → <code>202 {submissionId}</code> - Worker skeleton (<code>concurrency: 4</code>) emitting socket events via Redis adapter	- Problem Board: card grid, tier chips, difficulty filter - Problem Detail: react-markdown statement, constraints, examples - Resizable split layout (statement Monaco) via react-resizable-panels	- Seeded problems visible from real API - Submit → job picked up → dummy <code>grading:complete</code> received in browser
Day 5 Fri Jul 17	- GitHub Actions <code>ci.yml</code>: eslint + typecheck + prisma validate + build - Provision EC2 t3.medium; <code>docker-compose.prod.yml</code> - Caddyfile: auto-HTTPS <code>staging.talentforge.in</code>, proxy <code>/api</code> + <code>/socket.io</code>	- Wire auth flow against staging - Toasts (sonner), error boundary, 404 page - Loading skeletons; CI deploys frontend build to staging	WEEK 1 EXIT register → login → browse → editor, all green on staging - CI green on <code>main</code>; one-command deploy

WEEK 2 — Sandbox Execution & Auto-Grading (Jul 20–24) CRITICAL PATH

Day	Rohan — Backend / Sandbox / Infra	Karthikeyan — Frontend / Tools	Done When
Day 6 Mon Jul 20	- Runner images: <code>Dockerfile.python</code> (3.11-slim + pylint), <code>Dockerfile.node</code> (18-alpine + eslint), <code>Dockerfile.java</code> (temurin:17 + checkstyle) - Non-root user <code>runner</code>, entrypoint <code>run.sh</code> - dockerode HostConfig: <code>NetworkMode: 'none'</code>, <code>Memory: 256MB</code>, <code>ReadonlyRootfs: true</code>, tmpfs <code>/tmp 50m</code>, <code>PidsLimit: 64</code> 30s hard kill timer → status TIMEOUT	- Submit UX: button → “Queued...” spinner → live status chips (Queued → Running → Graded) <code>useGradingSocket(id)</code> hook: room <code>submission:{id}</code>, updates TanStack cache <code>ResultsPanel.tsx</code> collapsible shell under editor	- Hello-world runs in all 3 languages <code>curl</code> inside container fails (network off) - Infinite loop killed at 30s → TIMEOUT
Day 7 Tue Jul 21	<code>grader/correctness.ts</code>: pipe stdin per case, capture stdout/stderr/exit, weighted pass % → 0–100 - Per-case results persisted to <code>Submission.scores.cases[]</code> - Java: <code>javac</code> compile step; errors → COMPILE_ERROR + stderr	- Test-case table: pass/fail icons, expected-vs-actual diff (react-diff-viewer), time + memory per case - Animated SVG score ring (0→score) - State UIs: COMPILE_ERROR (mono stderr block), TIMEOUT, OOM	- Correct Two Sum = 100 in all 3 langs - Wrong solution shows exact failing cases + diff - Compile errors readable

Day	Rohan — Backend / Sandbox / Infra	Karthikeyan — Frontend / Tools	Done When
Day 8 Wed Jul 22	<code>grader/complexity.ts</code>: run at $n / 2n / 4n \rightarrow$ fit ratio $\rightarrow$ classify $O(n)/O(n \log n)/O(n^2)$ vs <code>expectedComplexity</code> <code>grader/style.ts</code>: pylint / eslint / checkstyle in-container $\rightarrow$ 0–100 - Composite <code>0.6·correctness + 0.3·complexity + 0.1·style</code>; emit in <code>grading:complete</code>	- Feedback panel: 3 score bars + composite - LLM feedback formatter: <code>POST /feedback/format</code> (Claude API proxy) $\rightarrow$ 3 coaching bullets, typing effect - Feature flag <code>VITE_FF_LLM_FEEDBACK</code>	$O(n^2)$ scores lower than $O(n)$ on complexity - Style violations reduce score - LLM bullets in <3s or flag off cleanly
Day 9 Thu Jul 23	<code>grader/precheck.ts</code> blocklist: py <code>subprocess os.system eval(,js child_process fs,java Runtime.getRuntime ProcessBuilder</code> $\rightarrow$ status BLOCKED pre-run - Graceful failures: queue never stalls; tuned <code>stalledInterval</code> - Retry: infra errors $\times 2$; user errors never	- Submission History page: attempts table (date, lang, score, status) $\rightarrow$ past-result detail - Score trend sparkline (Recharts LineChart) - Resubmit cooldown chip via server <code>nextAllowedAt</code>	- Fork bomb / fs read / net call rejected pre-execution - Worker kill mid-job $\rightarrow$ recovers, no zombie containers - History paginates correctly
Day 10 Fri Jul 24	Joint E2E: 10 canned solutions $\times$ 3 languages (correct / wrong / slow / malicious / infinite) — fix all - artillery load test: 20 concurrent submissions $\rightarrow$ p95 < 5s; tune concurrency, pre-pull images - Sentry (@sentry/node) in backend + worker	Joint E2E from browser; fix top 10 UX papercuts - Sentry React + session replay - Socket disconnect $\rightarrow$ auto-reconnect + 5s poll fallback	WEEK 2 EXIT submit $\rightarrow$ live score <5s p95, 3 langs, on staging 20 concurrent, 0 failures - Sentry captures test errors from both apps

WEEK 3 — Badges, Psychometric, Reviewer & Employer Portal (Jul 27–31)

Day	Rohan — Backend / Sandbox / Infra	Karthikeyan — Frontend / Tools	Done When
Day 11 Mon Jul 27	<code>services/badge.ts</code>: score $\geq 75 \rightarrow$ auto-create Badge (uuid <code>verifyId</code>, status <code>AI_VERIFIED</code>) - Public <code>GET /verify/:verifyId</code> (no auth) - Certificate: pdf-lib A4 + QR (qrcode) $\rightarrow$ S3 presigned download	- Badge celebration modal (canvas-confetti) on score ≥ 75 - Profile badge gallery: cards with status chip (AI / Expert verified) - Public <code>/verify/:id</code> page with OG tags + Download Certificate	- Score 80 $\rightarrow$ badge in gallery - Verify URL opens in incognito - PDF certificate QR resolves

Day	Rohan — Backend / Sandbox / Infra	Karthikeyan — Frontend / Tools	Done When
Day 12 Tue Jul 28	- Seed 40-item bank (30 Big-Five-lite Likert + 10 timed logic MCQ) <code>POST /psychometric/start / /submit</code> → traits: Logical 25 · Detail 20 · Learning 20 · Persistence 20 · Communication 15 - Gate problems list until profile exists	- Assessment UI: 15-min countdown, Likert + timed MCQs, progress bar - Autosave answers to sessionStorage - Results: Recharts RadarChart of 5 traits + interpretation copy	- New user routed to assessment first - Radar renders after submit - Refresh mid-test restores answers
Day 13 Wed Jul 29	- RBAC <code>requireRole('REVIEWER')</code>; seed reviewer account <code>GET /reviews/queue</code> (oldest AI_VERIFIED first); <code>POST /reviews/:id</code> {stars, comment, APPROVE/REJECT} - APPROVE → EXPERT_VERIFIED; REJECT → revoke + notify (notification table)	- Reviewer portal: queue list, read-only Monaco code viewer, star rating, comment, Approve/Reject - Student: notification bell (unread count), badge chip flips to Expert-verified, expert comment on result page	- Approve flips badge live - Reject shows reason to student - Queue oldest-first, empties correctly
Day 14 Thu Jul 30	<code>GET /candidates?minScore=&badge=&language=</code> with PG indexes on Badge.status, Submission.totalScore - Shortlist endpoints; code-sample access only if <code>profilePublic=true</code> - express-rate-limit + Redis: 100 req / 15 min on public routes	- Employer registration flow + Discover page: TanStack Table (sortable score, badge toggle, min-score slider, language filter) - Candidate drawer: scores, badges, psych radar, best code sample, Shortlist button - Shortlist page with remove	- Filter “score > 80 + expert-verified” returns correct list - Shortlist persists - Abuse test hits 429
Day 15 Fri Jul 31	- Seed to 8 problems incl. flagship “Build a Load Balancer” (10 hidden cases: round-robin, weighted, health-check) - Nightly cron: <code>pg_dump gzip aws s3 cp</code> (14-day retention) - Split envs: staging + <code>app.talentforge.in</code> prod	- LinkedIn share: OG image per badge + prefilled post - Landing page v1: hero, 3-step how-it-works, badge showcase, employer CTA - Onboarding tour (driver.js, 4 steps)	WEEK 3 EXIT full loop: register → psych → solve → badge → verify → employer shortlist 8 problems live; prod boots clean

WEEK 4 — Hardening, Beta & Blockchain Stretch (Aug 3–7)

Day	Rohan — Backend / Sandbox / Infra	Karthikeyan — Frontend / Tools	Done When
Day 16 Mon Aug 3	- helmet, strict CORS allowlist, zod audit on 100% endpoints - JWT rotation check; cookies <code>httpOnly · secure · sameSite=strict</code> <code>npm audit fix</code>; manual SQLi/XSS probes on search + comments	- Mobile pass: tables → cards <768px; editor read-only on mobile + “solve on desktop” banner - Sweep all routes: empty / error / loading states checklist	- ZAP baseline scan: no highs - Usable at 375px width - No blank screens on error
Day 17 Tue Aug 4	BETA LAUNCH invite 15–25 students - Live monitor: Sentry, <code>docker stats</code>, PG slow-query log - Same-day backend hotfixes + missing indexes	- Feedback channel (WhatsApp/Discord) + in-app feedback modal → <code>POST /feedback</code> - Same-day UI hotfixes; every report → GitHub issue (<code>beta</code> label)	≥15 registered, ≥10 graded submissions 0 open P1s end of day
Day 18 Wed Aug 5	- k6: 100 VUs (browse + submit mix) → fix bottlenecks - Compose <code>restart:always</code>, Redis AOF, disk-usage alert - Fix top-5 backend beta issues	Joint 1h feedback triage → ranked fixes; close top-5 UX issues - Lighthouse: perf > 90, LCP < 1.5s (React.lazy code-split, font preload)	- k6 100 VUs: error <1%, API p95 <200ms - Lighthouse ≥90 on Landing + Problems - Beta top-10 list closed
Day 19 Thu Aug 6	STRETCH Hardhat + OpenZeppelin <code>TalentForgeBadge.sol</code> (ERC721URIStorage, onlyOwner mint) - Deploy: <code>npm run hardhat run scripts/deploy.ts --network amoy</code> <code>POST /badges/:id/mint</code> server-side signer (ethers.Wallet) → store txHash + tokenId If behind: SKIP — DB badges ship	STRETCH “Mint on-chain” button → pending → success with PolygonScan link <code>amoy.polygonscan.com/tx/{hash}</code> - Hidden when <code>VITE_FF_CHAIN=false</code>	Badge mints on Amoy & scan link resolves — <i>or</i> flag off with zero side effects
Day 20 Fri Aug 7	<code>RUNBOOK.md</code>: queue stalled / DB down / sandbox breach / rollback - Backup restore drill on scratch DB Joint go/no-go review	- Copy edit sweep + FAQ page 2-min demo video; employer one-pager PDF Joint go/no-go review	WEEK 4 EXIT restore drill OK, runbook committed, go/no-go = GO, 0 P1/P2

Day	Rohan — Backend / Infra	Karthikeyan — Frontend / Comms	Done When
Day 21 Mon Aug 10	CODE FREEZE bugfix-only policy - Full prod regression: auth, submit ×3 langs, badge, verify, employer - DNS cutover + SSL check <code>app.talentforge.in</code>	- Prod regression on Chrome / Firefox / Safari + 2 phones - OG previews validated (LinkedIn Post Inspector)	Regression checklist 100% green on production
Day 22 Tue Aug 11	SOFT LAUNCH ~50 invited students - Watch Sentry + BullMQ queue depth all day	- Send invites; monitor first real sessions (session replay) - Same-day fixes for onboarding friction	≥30 real signups, ≥15 graded submissions, 0 P1s
Day 23 Wed Aug 12	- Fix soft-launch findings - UptimeRobot monitors: <code>/health</code>, frontend, verify URL	- Employer outreach: one-pager to 5–10 HR contacts; assist account creation - Collect 2–3 student quotes for launch post	≥3 employer accounts, ≥1 shortlist made
Day 24 Thu Aug 13	- Final DB backup + snapshot; alert thresholds confirmed - On-call plan for launch day agreed	Launch assets final: LinkedIn founder post, college groups, Discord announcement, demo video pinned	Posts drafted & scheduled; backups verified
Day 25 Fri Aug 14 	PUBLIC LAUNCH on-call all day: grading success rate, uptime, queue - Evening: metrics snapshot for investor / customer deck	- Publish posts 10:00 IST; respond in channels; hotfix UI - Evening: “first badges earned” follow-up post	Targets: 100+ signups · 30+ submissions · 15+ badges · 3–5 employers · 1+ shortlist · 99%+ uptime · grading p95 <5s

Daily rhythm: 9:30 AM IST standup (15 min) · all work via PRs, cross-reviewed within 12h · Friday 4 PM demo + retro · staging auto-deploys on merge to `main`, prod deploys manual & tagged. **Rule of the month:** ECE, tiers, matching engine, MOSS and payments are post-launch — say no until Aug 15.